

PCB Technology Specification



Document Number d019762_07_S01
Project Name ek-ra8d2_mp
Variant None
Version 0301
Date 11.06.2025

General requirements

IPC-A-600	Class II
IPC-A-6011/IPC-A-6012	Class II
Number of Layers	see drawing
Conductor min Width / Spacing	60 µm
Finished Board Thickness	1.40 mm
Pad finish	chem. Ni/AU (ENIG)
Impedance Check	yes
E-Test	yes
First sample test report	no
Finished copper inside	35µm
Finished copper outside	35µm

Materials and Ratings

Base Material	FR 4
Tg (°C min)	170 °C
Td (°C min)	330 °C
RoHS compatible	yes
UL94V Rating	UL94V-0

Drill and Milling

Smallest drill diameter	100 µm
Smallest drill pad	220 µm
Backdrill	no
Blind via	no
Buried via	no
Microfill blind via	no
Plugging buried via	no
Via Processing	IPC 4761 Type VII
Milling tolerance	± 100 µm

Production requirements

PCB-Type	Rigid
Finished PCB Dimension (mm)	84 mm x 190 mm
Create Panel	no 5 x 5
Panel Dimension (mm)	525 mm x 500 mm
Contour processing	Milling
Add tooling holes in panel	no
Inner Thieving Balancing Allowed	no
Outer Thieving Balancing Allowed	no
Gold Fingers	no
Edge contact	no
Soldermask Product	KSM-6189 BLO3
Soldermask Color (Top & Bottom)	Renesas Blue CMYK(73,73,0,38 RGB (42,40,157)
Soldermask finish	semi-gloss
Silkscreen	Top & Bottom
Silkscreen Color	white
Silkscreen printing method	Silkscreen Printed

Add additional marking

Board Lot #	no
QC Stamp	no
UL94V	no
RoHS	no
Manufacturer Logo	no
Date code	no

Comment

DNF components are not soldered and so pads for these components must be free from solder. No changes are to be made without approval from Renesas. This includes the addition of circuit fiducials but excludes etch compensation.
This PCB is not to be marked with UL Logos but processed in a UL Approved process.